

Multiple Choice (10 marks)

1. Statement 1 | Every integral domain has a field of quotients. Statement 2 | A polynomial of degree n over a ring can have at most n zeros counting multiplicity.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1 | The homomorphic image of a cyclic group is cyclic. Statement 2 | The homomorphic image of an Abelian group is Abelian.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. For which value of n are there exactly two abelian groups of order n up to isomorphism?
 - (a) 4
 - (b) 7
 - (c) 8
 - (d) 12
4. Let $f(x, y) = x^3 + y^3 + 3xy$ for all real x and y . Then there exist distinct points P and Q such that f has a
 - (a) local maximum at P and at Q
 - (b) saddle point at P and at Q
 - (c) local maximum at P and a saddle point at Q
 - (d) local minimum at P and a saddle point at Q
5. Find the maximum possible order for an element of S_n for $n = 6$.
 - (a) 6
 - (b) 12
 - (c) 30
 - (d) 105

True / False (10 marks)

Answer True or False

6. True or False: The polynomial $4x^2 - 2$ is irreducible over the integers $\mathbb{Z}$.
7. True or False: If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$.
8. True or False: The area of an equilateral triangle with an inscribed circle of radius 2 is 16.
9. True or False: There are exactly two homomorphisms from the group of integers $\mathbb{Z}$ into the group $\mathbb{Z}_2$.
10. True or False: The integers n such that $3 \leq n \leq 11$ with only one group of order n (up to isomorphism) are 3, 5, 7, and 11.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the given system of linear equations and determine which statements about its solutions are true or false. Discuss the implications of your findings regarding the solution's form.
12. Discuss the implications of two similar $n \times n$ invertible matrices A and B , specifically addressing the similarity of the matrices $A - 2I$, the preservation of trace, and the similarity of their inverses.

End of Paper